

Linear Algebra

Linear Combination:

It means multiplying variables (or vectors) by numbers and then adding them together.

Example:-

$$\vec{v}_1 = (1, 2)$$

$$\vec{v}_2 = (3, 4)$$

Linear combination would be:

$$2\vec{v}_1 + 3\vec{v}_2$$

Now:

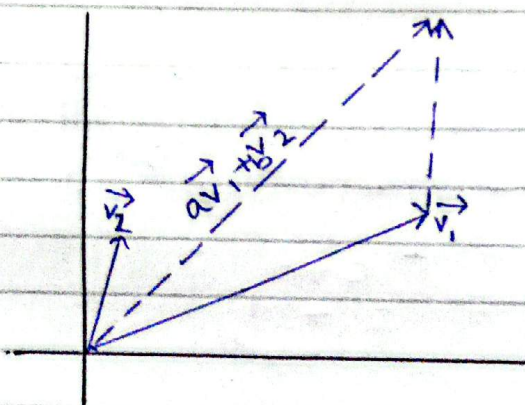
$$2(1, 2) + 3(3, 4)$$

$$(2, 4) + (9, 12)$$

$$(11, 16)$$

So $(11, 16)$ is the Linear combination of $\vec{v}_1$ and $\vec{v}_2$

↳ This also lie in same plane as $(1, 2)$ and $(3, 4)$



Questions:-

$$\vec{u} = (1, 2) \text{ or } \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\vec{v} = (4, 5) \text{ or } \begin{bmatrix} 4 \\ 5 \end{bmatrix}$$

$$a\vec{u} + b\vec{v} = \vec{w}$$

Q:- $\vec{w} = (7, 8)$

with the linear combination of $\vec{u}$ and $\vec{v}$ can we get $\vec{w} = (7, 8)$

Que: 2 $w = (7, 8) = a\vec{u} + b\vec{v} = ?$

Key Point:- If we take the Linear combination of any two vectors the resultant vector lie in same plane as those two vectors.

Example (In 3D):-

$$\vec{v} = (1, 2, 3)$$

$$\vec{u} = (1, 3, 4)$$

$$\text{matrix} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix}$$

Linear combination of $\vec{u}$ and $\vec{v}$ will be

$$c\vec{v} + d\vec{u}$$

$$c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + d \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$$

Verify:

$$c \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + d \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix}$$

$$\begin{bmatrix} c \\ 2c \\ 3c \end{bmatrix} + \begin{bmatrix} d \\ 3d \\ 4d \end{bmatrix} = \begin{bmatrix} c+d \\ 2c+3d \\ 3c+4d \end{bmatrix}$$

$$\begin{bmatrix} c+d \\ 2c+3d \\ 3c+4d \end{bmatrix} = \begin{bmatrix} c+d \\ 2c+3d \\ 3c+4d \end{bmatrix}$$

Questions:-

1- Is $b = (2, 5, 7)$ lies in same plane as $\vec{v}$ and $\vec{u}$
 $\vec{v} = (1, 2, 3)$ and $\vec{u} = (1, 3, 4)$

2- with the linear combination of $\vec{u}$ and $\vec{v}$ can we get $\vec{b}$.

3- Does $\begin{bmatrix} 1 & 1 \\ 2 & 3 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}$ solution exist?

4- Does $Ax = b$ has solution?

$\downarrow$ $\downarrow$
Coefficient Unknown
Matrix

(yes exist if columns of A are independent).

Example:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 4 \end{bmatrix}$$

A Does't have solution because

$$c_1 + c_2 = c_3$$

$$c_3 - c_1 = c_2$$

$$c_3 - c_2 = c_1$$

Key Terms:-

1. Column Space:- ALL combinations of column.

2. Row Space: ALL linear combination of Row.

3. Rank: No of independent column.

4. Elimination: Good way to find Rank.

System of Linear Equations

Finite set of Linear Equations.

$$ax + by = c \quad (a \neq 0, b \neq 0)$$
$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = b$$

Homogenous Linear Equation.

where right hand side is 0.

$$a_1x_1 + a_2x_2 + a_3x_3 + \dots + a_nx_n = 0$$

Identify Linear Eqs:

$$x + 3y^2 = 4$$

$$3x + xy - xy = 5 \quad \times$$

$$\sqrt{x} + 2y = 5 \quad \times$$

$$\sin x + y = 7 \quad \times$$

$$e^x + y = 9 \quad \times$$

$$x + y = \sin 90^\circ \quad \checkmark$$

$$\left(\cos \frac{\pi}{5}\right)x + 9y = 8 \quad \checkmark$$

Key Point: where the two lines intersect that's called the solution of linear equations.

Example:-

$$5x + y = 3$$

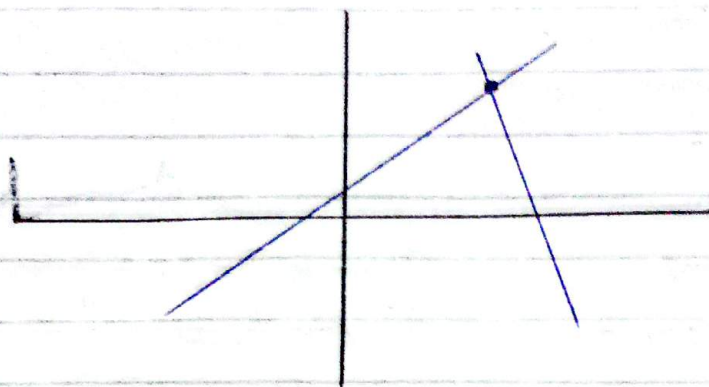
$$2x - y = 4$$

(1, -2) is the solution of above system.

1. Linear System with one solution:-

The lines may intersect at only one point in which case the system has exactly one solution.

Geometry:-



Example: 1

$$\begin{aligned}x - y &= 1 \rightarrow 1 \\2x + y &= 6 \rightarrow 2\end{aligned}$$

$$\begin{aligned}\text{x-ply eq (1) by 2} \\2(x - y = 1) \\2x - 2y = 2 \rightarrow (3)\end{aligned}$$

$$\begin{aligned}\text{Subs eq 3 From 2} \\+ 2x + y = +6 \\- \pm 2x \mp 2y = \pm 2 \\3y = 4 \\y = \frac{4}{3}\end{aligned}$$

$$\text{put } y = \frac{4}{3} \text{ in (1)}$$

$$x - \frac{4}{3} = 1$$

$$x = 1 + \frac{4}{3} \Rightarrow x = \frac{3+4}{3}$$

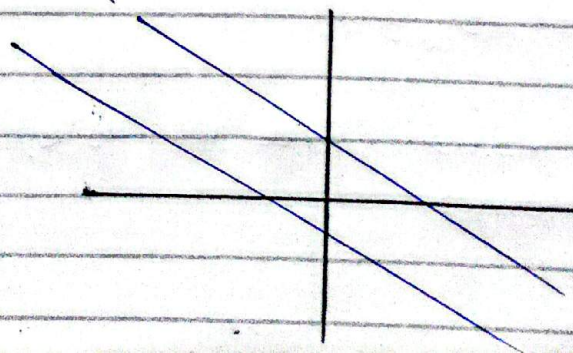
$$x = \frac{7}{3}$$

$(\frac{7}{3}, \frac{4}{3}) \rightarrow$ One solution.

2. Linear System with no solution:

The lines may be parallel and distinct in which case there is no solution (coz no intersection).

Geometry:-



Example:-

$$x + y = 4 \rightarrow (1)$$

$$3x + 3y = 6 \rightarrow (2)$$

x-PLY eq (1) with 3

$$3(x + y = 4)$$

$$3x + 3y = 12 \rightarrow 3$$

subs eq 3 From eq 2

$$+ \quad 3x + 3y = +6$$

$$- \quad 3x + 3y = +12$$

$$0 = -6$$

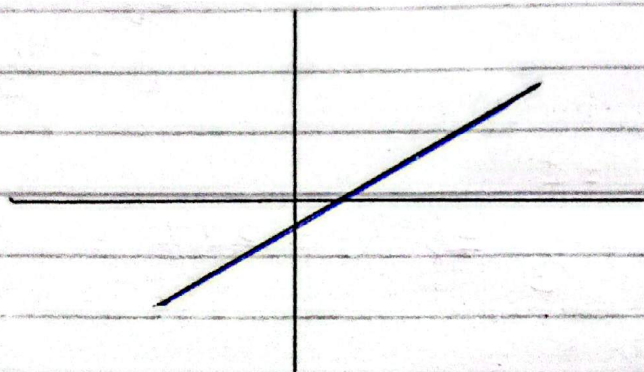


No solution (coz not possible)

3. Linear System with Infinite Solution:

The Line may coincide in which there are many solution.

Geometry:-



Example:-

$$4x - 2y = 1 \rightarrow 1$$

$$16x - 8y = 4 \rightarrow 2$$

These both eq are same
just x-PLY eq (1) with 4

Example: 2

$$x - y + 2z = 5 \rightarrow 1$$

$$2x - 2y + 4z = 10 \rightarrow 2$$

$$3x - 3y + 6z = 15 \rightarrow 3$$

Let $z = t$, $y = w$ (Free variable)

So eq (1) becomes

$$x = 5 + w - 2t$$

If $w=0$ and $t=0$

$$x = 5 + 0 - 2(0)$$

$$x = 5$$

$$y = 0$$

$$(5, 0)$$

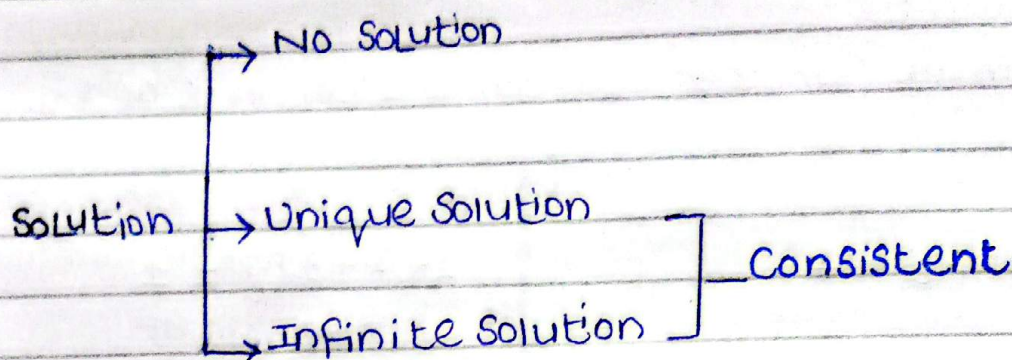
If $w=1$ and $t=0$

$$x = 5 + 1 - 2(0)$$

$$x = 6$$

$$y = -10$$

$$(6, -10)$$



Exercise : 1.1

Question : 01

Determine whether equation is linear in x_1, x_2 and x_3 .

a-

$$x_1 + 5x_2 - \sqrt{2}x_3 = 1$$

↳ Linear

d-

$$x_1^{-2} + x_2 + 8x_3 = 5$$

↳ Non-Linear.

• All variable have power 1

• $\sqrt{2}$ is constant

$$x_1^{-2} = \frac{1}{x_1^2}$$

• Variable has power -2.

b-

$$x_1 + 3x_2 + x_1x_3 = 2$$

↳ Not linear

e-

$$x_1^{3/5} - 2x_2 + x_3 = 4$$

↳ Non Linear.

• Product (x_1, x_3) exist.

$x_1^{3/5}$ → Fractional Power.

$$x_1 = -7x_2 + 3x_3$$

↳ Linear

- ALL variables power 1
- NO product exist

c-

$$\pi x_1 - \sqrt{2} x_2 = 7^{\frac{1}{3}}$$

↳ Linear

- variables have power 1
- $\pi, \sqrt{2}, 7^{\frac{1}{3}}$ are constant

Question: 2

Determine whether equation is Linear in x and y .

a-

$$2^{\frac{1}{3}}x + \sqrt{3}y = 1$$

↳ Linear

- $2^{\frac{1}{3}}$ and $\sqrt{3}$ are constant
- variables have power 1.

c-

$$\cos\left(\frac{\pi}{7}\right)x - 4y = \log 3$$

↳ Linear

- $\cos\left(\frac{\pi}{7}\right)$ and $\log 3$ constant
- variables have power 1.

e-

$$xy = 1$$

↳ Non Linear

- Product exist

b-

$$2x^{\frac{1}{3}} + 3\sqrt{y} = 1$$

↳ Non Linear

- $x^{\frac{1}{3}}$ Fractional Power
- $\sqrt{y}$ power $\frac{1}{2}$

d-

$$\frac{\pi}{7} \cos x - 4y = 0$$

↳ Non Linear

- x is in trig Function.
- So Non Linear operation.

f-

$$y + 7 = x$$

↳ Linear

- No Product
- variables have power 1.

Question: 09

Determine whether the given 3-Tuple is a Solution of Linear System.

$$2x_1 - 4x_2 - x_3 = 1$$

$$x_1 - 3x_2 + x_3 = 1$$

$$3x_1 - 5x_2 - 3x_3 = 1$$

a-

$(3, 1, 1)$

Eq: 1

$$2(3) - 4(1) - 1 = 1$$

$$6 - 4 - 1 = 1$$

$$1 = 1$$

Eq: 2

$$2 - 3(1) + 1 = 1$$

$$2 - 3 + 1 = 1$$

$$1 = 1$$

Eq: 3

$$3(3) - 5(1) - 3(1) = 1$$

$$9 - 5 - 3 = 1$$

$$1 = 1$$

Solution: Satisfy all equations.

b-

$(3, -1, 1)$

Eq: 1

$$2(3) - 4(-1) - 1 = 1$$

$$6 + 4 - 1 = 1$$

$$9 \neq 1$$

Eq: 2

$$3 - 3(-1) + 1 = 1$$

$$3 + 3 + 1 = 1$$

$$7 \neq 1$$

Eq: 3

$$3(3) - 5(-1) - 3(1) = 1$$

$$9 + 5 - 3 = 1$$

$$11 \neq 1$$

Not a Solution

C-

$$(13, 5, 2)$$

Eq. 1:-

$$2(13) - 4(5) - 2 = 1$$

$$26 - 20 - 2 = 1$$

$$4 \neq 1$$

Eq. 2:-

$$13 - 3(5) + 2 = 1$$

$$13 - 15 + 2 = 1$$

$$0 \neq 1$$

Eq. 3

$$3(13) - 5(5) - 3(2) = 1$$

$$39 - 25 - 6 = 1$$

$$9 \neq 1$$

Does't

Not Solution: [↑] Satisfy Any Equation

d

$$(13\frac{1}{2}, 5\frac{1}{2}, 2)$$

Eq. 1

$$2(13\frac{1}{2}) - 4(5\frac{1}{2}) + 2 = 1$$

$$13 - 10 - 2 = 1$$

$$1 = 1$$

Eq. 2

$$13\frac{1}{2} - 3(5\frac{1}{2}) + 2 = 1$$

$$13\frac{1}{2} - 15\frac{1}{2} + 2 = 1$$

$$\frac{13 - 15 + 4}{2} = 1$$

$$1 = 1$$

Eq. 3

$$3(13\frac{1}{2}) - 5(5\frac{1}{2}) - 3(2) = 1$$

$$39\frac{1}{2} - 25\frac{1}{2} - 6 = 1$$

$$\frac{39 - 25 - 12}{2} = 1$$

$$1 = 1$$

Solution: Satisfy ALL equations.

d-

$$(17, 7, 5)$$

Eq: 1

$$2(17) - 4(7) - 5 = 1$$

$$34 - 28 - 5 = 1$$

$$1 = 1$$

Eq: 2

$$17 - 3(7) + 5 = 1$$

$$17 - 21 + 5 = 1$$

$$1 = 1$$

Eq: 3

$$3(17) - 5(7) - 3(5) = 1$$

$$51 - 35 - 15 = 1$$

$$1 = 1$$

Solution:- Satisfy ALL equations.

Question: 10-

$$x + 2y - 2z = 3$$

$$3x - y + z = 1$$

$$-x + 5y - 5z = 5$$

a-

$$(5/7, 8/7, 1)$$

Eq: 1

$$5/7 + 2(8/7) - 2(1) = 3$$

$$\frac{5 + 16 - 7}{7} = 3$$

$$1 \neq 3$$

Eq: 2

$$3(5/7) - 8/7 + 1 = 1$$

$$\frac{15}{7} - 8/7 + 1 = 1$$

$$\frac{15 - 8 + 7}{7} = 1$$

$$2 \neq 1$$

Not Solution: Not a Solution bcz doesn't satisfy 1st two equations.

b-

$$(5/7, 8/7, 0)$$

Eq: 1

$$5/7 + 2(8/7) - 2(0) = 3$$

$$5/7 + 16/7 - 0 = 3$$

$$\frac{21}{7} = 3$$

$$3 = 3$$

Eq: 2

$$3(5/7) - 8/7 + 0 = 1$$

$$\frac{15 - 8}{7} = 1$$

$$7/7 = 1$$

$$1 = 1$$

Eq: 3

$$-5/7 + 5(8/7) - 5(0) = 5$$

$$\frac{-5 + 40}{7} = 5$$

$$5 = 5$$

Solution:

c-

$$(5, 8, 1)$$

Eq: 1

$$5 + 2(8) - 2(1) = 3$$

$$5 + 16 - 2 = 3$$

$$19 \neq 3$$

Not Solution: Does't Satisfy
1st Equation.

d-

$$(5/7, 10/7, 2/7)$$

Eq: 1

$$5/7 + 2(10/7) - 2(2/7) = 3$$

$$\frac{5 + 20 - 4}{7} = 3$$

$$3 = 3$$

Eq: 2

$$3(5/7) - 10/7 + 2/7 = 1$$

$$\frac{15 - 10 + 2}{7} = 1$$

$$1 = 1$$

Eq: 3

$$\begin{aligned} -5/7 + 5(10/7) - 5(2/7) &= 5 \\ \frac{-5 + 50 - 10}{7} &= 5 \\ 5 &= 5 \end{aligned}$$

Solution: satisfy ALL equations.

e-

$$(5/7, 22/7, 2)$$

Eq: 1

$$\begin{aligned} 5/7 + 2(22/7) - 2(2) &= 3 \\ \frac{-5 + 44 - 28}{7} &= 3 \\ 3 &= 3 \end{aligned}$$

Eq: 2

$$\begin{aligned} 3(5/7) - 22/7 + 2 &= 1 \\ \frac{15 - 22 + 14}{7} &= 1 \\ 1 &= 1 \end{aligned}$$

Eq: 3

$$\begin{aligned} -5/7 + 5(22/7) - 5(2) &= 5 \\ \frac{-5 + 110 - 10}{7} &= 5 \\ 5 &= 5 \end{aligned}$$

Solution: satisfy all equations.

Question: 11

Each part Determine whether unique, no or infinite Solution.

a-

$$\begin{aligned} 3x - 2y &= 4 \rightarrow (1) \\ 6x - 4y &= 9 \rightarrow (2) \end{aligned}$$

x-PLY eqn (1) by (2)

$$2(3x - 2y = 4)$$

$$6x - 4y = 8 \rightarrow (3)$$

Subtract eq (3) and eq (2)

$$\begin{array}{r} + 6x - 4y = 9 \\ - \pm 6x \mp 4y = \pm 8 \\ \hline \end{array}$$

$$0 = 1 \rightarrow \text{Contradiction}$$

$\hookrightarrow$ NO SOLUTION. (Lines are ||)

b-

$$2x - 4y = 1 \rightarrow (1)$$

$$4x - 8y = 2 \rightarrow (2)$$

x-PLY eq (1) by 2. subtract eq (2) and eq (3)

$$2(2x - 4y = 1)$$

$$4x - 8y = 2 \rightarrow (3)$$

$$+ 4x - 8y = +2$$

$$- \pm 4x \mp 8y = \pm 2$$

$$0 = 0$$

$\hookrightarrow$ Infinitely many solutions.

2nd way:-

both eq represent the same line.

From:-

$$2x - 4y = 1$$

$$2x = 1 + 4y$$

$$x = \frac{1}{2} + 2y$$

Let $y = t$

$$\boxed{x = \frac{1}{2} + 2t}$$

$$\boxed{y = t}$$

where $t \in \mathbb{R}$.

c-

$$x - 2y = 0 \rightarrow (1)$$

$$x - 4y = 8 \rightarrow (2)$$

Subtract eq (1) and eq (2)

$$+ x - 2y = +0$$

$$\pm x \mp 4y = \pm 8$$

$$2y = -8$$

$$\boxed{y = -4}$$

put $y = -4$ in eq (1)

$$\boxed{x = -8}$$

$\hookrightarrow$ Unique solution.

Question: 12

Under what condition on a and b will following Linear system have no solutions, one solution or infinitely many solutions?

$$2x - 3y = a \rightarrow (1)$$

$$4x - by = b \rightarrow (2)$$

x-PLY eq 1 by 2

$$2(2x - 3y = a)$$

$$4x - 6y = 2a$$

$$4x - by = 2a$$

$$4x - by = b$$

↳ No solution if $2a \neq b$

↳ Infinite Solution if $2a = b$

↳ Can't have unique solution.

Question: 19

Find all values of k for which given augmented matrix is consistent.

a-

$$\begin{bmatrix} 1 & k & -4 \\ 4 & 8 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & k & -4 \\ 0 & 8-4k & 18 \end{bmatrix}$$

$$R_2 - 4R_1 \rightarrow x_1 + kx_2 = 4 \rightarrow (1)$$

$$0x_1 + (8-4k)x_2 = 18 \rightarrow (2)$$

Check For Consistency:-

From eq (2)

$$(8-4k)x_2 = 18$$

If $k = 2$

$$0 \neq 18$$

↳ Contradiction.

Conclusion:-

↳ System is inconsistent for $k = 2$

↳ For system to be consistent $k \neq 2$ or $k \in \mathbb{R} - \{2\}$

b-

$$\begin{bmatrix} 1 & k-1 \\ 4 & 8 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & k-1 \\ 0 & 8-4k & 0 \end{bmatrix} \xrightarrow{R_2 - 4R_1} \begin{aligned} x_1 + kx_2 &= -1 \rightarrow (1) \\ 0x_1 + (8-4k)x_2 &= 0 \rightarrow (2) \end{aligned}$$

check for consistency

From eq (2)

$$(8-4k)x_2 = 0$$

If $k = 2$

$$0 = 0$$

Conclusion: system is consistent
For all values of k

$$k \in \mathbb{R}$$

Question: 20

a

$$\begin{bmatrix} 3 & -4 & k \\ -6 & 8 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -4 & k \\ 0 & 0 & 5+2k \end{bmatrix} \xrightarrow{R_2 + 2R_1} \begin{aligned} 3x_1 - 4x_2 &= k \rightarrow (1) \\ 0x_1 + 0x_2 &= 5+2k \rightarrow (2) \end{aligned}$$

check for consistency

From eq (2)

$$0 = 5+2k$$

If $k = -5/2$

$$0 = 5+2(-5/2)$$

$$0 = 0$$

Conclusion: For a system to be consistent k must be

equal to $-5/2$

$$k \in -5/2$$

b-

$$\begin{bmatrix} k & 1 & -2 \\ 4 & -1 & 2 \end{bmatrix}$$

$$\begin{bmatrix} k & 1 & -2 \\ 4+k & 0 & 0 \end{bmatrix} \xrightarrow{R_2 + R_1} \begin{aligned} kx_1 + x_2 &= -2 \rightarrow (1) \\ (4+k)x_1 + 0x_2 &= 0 \rightarrow (2) \end{aligned}$$

check For consistency

From eq (2)

$$(4+k)x_1 = 0 \rightarrow$$

$$\text{If } k = -4$$

$$0 = 0$$

Conclusion: System is consistent
For all values of k

$$k \in \mathbb{R}$$
